APPENDIX I: NEPAL—DEBT SUSTAINABILITY ANALYSIS

Public debt dynamics are assessed using the Low-Income Country Debt Sustainability Analysis (LIC-DSA) framework, which was jointly prepared with the World Bank. Based on the external LIC-DSA, Nepal's external debt dynamics are subject to a high risk of debt distress. This assessment is broadly the same as in the LIC-DSA conducted in 2006; the baseline scenario is also roughly the same as in the previous DSA. The initial net present value of debt is slightly lower than in the previous DSA due to lower than projected disbursements in the interim.

VI. LOW-INCOME COUNTRY DEBT SUSTAINABILITY ANALYSIS (LIC-DSA)[1]

A. Recent Debt Developments

1. **Nepal's total debt stock is estimated at 59½ percent of GDP at end-2005 (in nominal terms)**. External debt is estimated at US$3½ billion (46 percent of GDP) of which about US$3 billion was owed to multilateral institutions, mostly IDA and the AsDB. Bilateral debt stock is estimated at about US$270 million, with Japan being the largest creditors accounting for more than half of the bilateral debt. After remaining fairly constant at a little over 50 percent as a share of GDP since 1995, the external debt stock as share of GDP dipped somewhat in 2005. External loan disbursements in recent years have been relatively low. Domestic debt also declined somewhat in 2005 by about 1 percent of GDP.

B. Assumptions

2. **Baseline projections of PPG debt are based on the following key assumptions**:

- **Real sector**: Real GDP growth is projected to rise gradually from 2 percent in 2005/06 and stabilize at 5½ percent after 2009/10, in line with growth rates observed in the early 1990s and supported by structural reforms and sound macroeconomic policies. Inflation is assumed to decline from about 8 percent in 2005/06 to 5 percent in the medium term as supply bottlenecks are gradually alleviated. The nominal interest rate on domestic debt is expected to average 6¼ percent during 2006/07−2009/10 and 6 percent thereafter. The exchange rate is projected to depreciate against the dollar, in line with projected movements in the Indian rupee to which the Nepalese rupee is pegged.

- **Fiscal sector**: The revenue-to-GDP ratio is projected to rise from 12¼ percent in 2005/06 to 13½ percent by 2008/09, assuming a rebound in economic activity and

[1] Following *Operational Framework for Debt Sustainability Assessments in Low-Income Countries—Further Considerations* (SM/05/109 and IDA/R2005–0056).

improved tax collection from a broadening of the tax base. The expenditure-to-GDP ratio rises from 16½ percent in 2005/06 to 19½ percent in 2008/09 as development spending picks up, and stabilizes thereafter. Growth of real primary spending averages 7½ percent in 2006/07−2010/11 (5½ percent in 2011/12−2025/26), with domestic financing of 1–1⅓ percent of GDP and official grants around 3 percent of GDP in 2006/07–2010/11(around 1½ percent of GDP in 2011/12−2025/26).

- **External sector**: Exports of goods and services are projected to grow at an average 8½ percent over the projection period supported by tourism and strong growth in partner countries. Imports of goods and services are expected to rebound in line with economic activity, with growth averaging 7 percent. The noninterest current account balance is projected to deteriorate from a surplus of 2½ percent of GDP in 2004/05 to a deficit of 2¼ percent of GDP by 2025/26, with remittances declining gradually as a share of GDP. Scheduled debt service on existing external debt increases from around US$110 million in 2005/06 to US$125 million in 2025/26. New financing is assumed to be moderate during the next three years, but rising gradually from about US$200 million in 2006/07 to US$350 million by 2009/10. The grant element of new borrowing is assumed to average about 44 percent in 2006/07 and about 40 percent in the projection period.

C. Baseline

PPG External Debt

3. **A key feature of the LIC−DSA framework is that it compares debt burden indicators to indicative policy-based thresholds**. The thresholds are based on the empirical finding that low-income countries with stronger policies and institutions tend to have a higher debt carrying capacity.[2] Nepal is classified as a medium performer. At end 2006, Nepal's NPV of public debt-to-exports ratio is estimated at 163 percent (the relevant policy based indicative threshold is 150 percent). The ratio is projected to fall to 137 percent by 2010/11 and 87 percent by 2025/26. Other indicators remain below the policy-based indicative thresholds throughout the projection period.

Nepal: Indicative External Debt Burden Indicators

	Indicative Thresholds 1/	Nepal 2004/05 2/	Nepal 2005/06	Nepal Average 2006/07–2025/26
NPV of debt, in percent of				
Exports	150	172	163	120
GDP	40	28	25	20
Revenues	250	213	211	164
Debt service, in percent of				
Exports	20	9	9	9
Revenues	30	12	12	13

1/ Shows indicative policy dependent thresholds under the joint IMF-World Bank low-income country debt sustainability framework for a medium policy performer.
2. Appendix II, IMF Country Report No. 06/44, *Nepal–Staff Report for the 2005 Article IV Consultation*.

[2] See IDA/R2005–0056 and SM/05/109.

4. **In the baseline scenario, debt burden ratios are projected to fall between 2005/06 and 2025/26** (Table I.1). The NPV of external public debt-to-GDP (25 percent to 16 percent) NPV of external public debt-to-exports (163 percent to 87 percent); external public debt service-to-exports ratio (9 percent to 8 percent).[3]

Total Public Debt

5. **Domestic debt accounts for about 28½ percent of total public debt**. For the baseline scenario, the NPV of public debt-to-GDP ratio declines from 44 percent at end 2005/06 to 34 percent by 2025/26 (Table I.4, Figure I.2). Over the same period, the NPV of public debt to-revenue ratio falls from 303 percent to 209 percent, and the public debt service-to revenue ratio decreases from 25 percent to 16 percent.

D. Sensitivity Analysis

6. **Stress tests and alternative scenarios suggest susceptibility to shocks**.

- **Total public debt**: Following an extreme shock—a one-time 30 percent real depreciation in 2006/07 which leads to the highest level of NPV of debt-to-GDP 10 years after the shock—the NPV ratio peaks at around 55 percent in 2006/07 and falls to about 40 percent in 2025/26. The debt service to revenue ratio peaks in 2005/06 at 25 percent declining to 19 percent in 2025/26. These scenarios are especially sensitive to GDP projections, including the prospects for the peace process in the near term. Over the medium term, the effectiveness of public investment the availability of external financing, and fiscal risks such as those from contingent liabilities in the financial and public enterprise sectors are key.[4]

- **External debt**: Following an export shock—defined as export value growth at historical levels minus one standard deviation in the first two years of the shock—the NPV of debt-to-exports ratio increases significantly, peaking at a little over 330 percent in the near term, but quickly dissipates thereafter. The debt service-to-exports ratio peaks at 19 percent in 2015/16. These results are driven by Nepal's volatile export performance in the historical period. The combined shocks scenario has a similar effect, primarily based on the export component of the shock. If financing is available only at less favorable terms, debt indicators would remain

[3] Given the high concessionality of external debt, the debt service-to-exports ratio is low, and at levels similar or lower than to most HIPCs after full HIPC debt relief. The ratio reflects debt service on existing debt and debt service on projected disbursements.

[4] Liabilities from financial sector and public enterprise reforms are expected to be 7–8 percent of GDP, with annual carrying costs of ¼–¾ percent of GDP.

higher than in the baseline and the debt service ratio would start to rise. While debt indicators go below the baseline in the historical scenario toward the end of the projections, it reflects the effect of extremely low current account deficits and even surpluses in the historical period that cannot be expected to persist.[5]

E. Staff Assessment

7. **Based on the LIC-DSA, staffs conclude that Nepal's external debt dynamics are subject to a high risk of distress**. The external debt-to-export ratio is above the indicative threshold in 2005 and could remain so under various scenarios, especially if growth is low, donor grant support inadequate, and contingent liabilities high. The sensitivity analyses underscore the need to implement sound macroeconomic policies and reforms, including toward achieving higher export growth, while maintaining efforts to raise revenue. These factors combined with foreign financing at favorable terms—preferably through grants—would help Nepal make progress toward achieving its MDG targets while containing risks to debt sustainability.

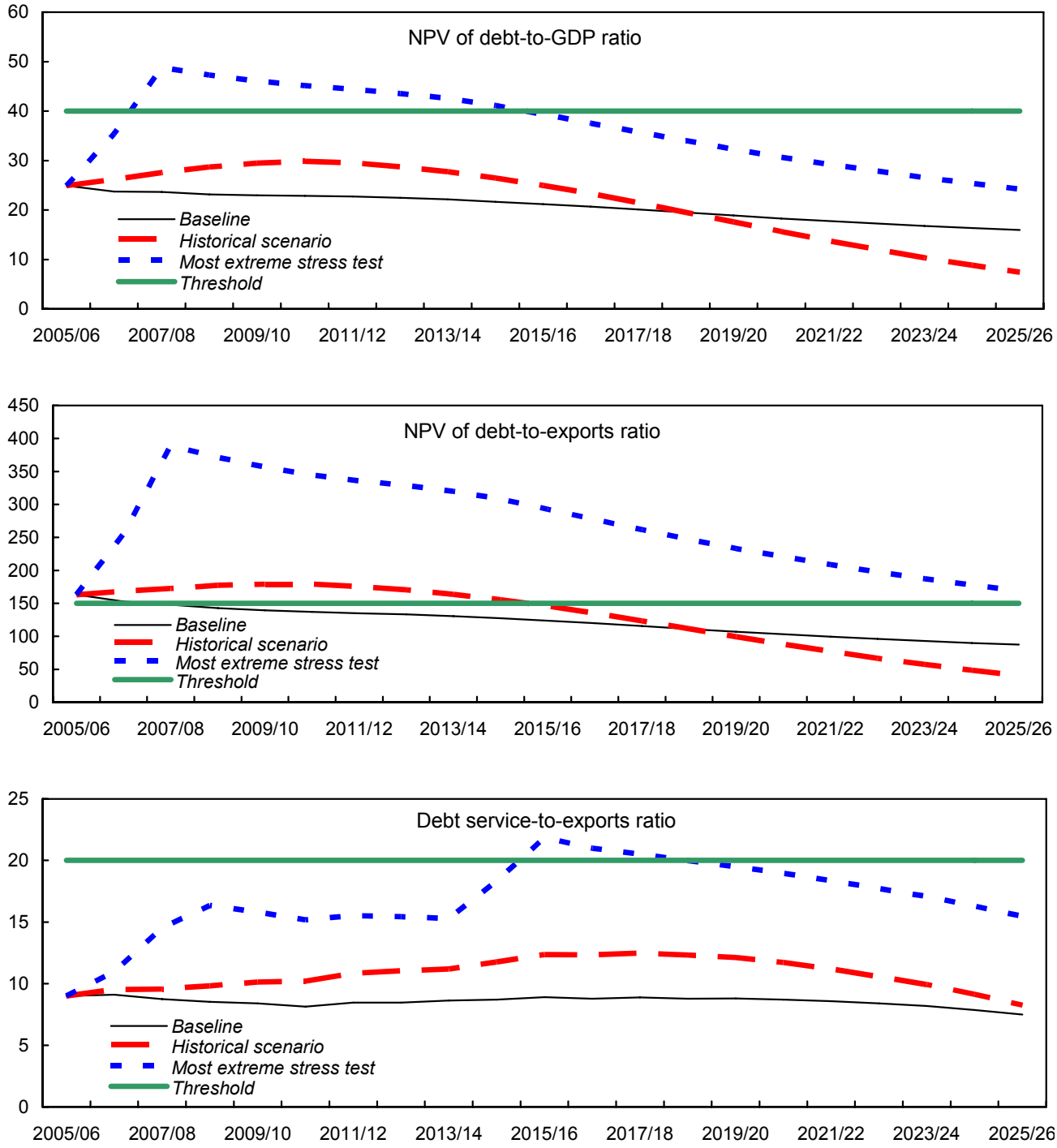
VII. Enhanced HIPC and Multilateral Debt Relief (MDR) Initiatives

8. **Nepal was recently deemed potentially eligible for HIPC relief under the extended sunset clause**. This determination was based on an estimated end 2004 NPV of debt-to-exports ratio (around 200 percent after traditional debt relief; the enhanced HIPC threshold is 150 percent). If Nepal's eligibility is confirmed at the appropriate time, it could receive substantial relief under the enhanced HIPC Initiative and MDRI.

[5] In the baseline projections, current account surpluses are initially larger than the average in the historical scenario. The historical scenario uses average of the late 1990s, a more appropriate reference period.

Figure I.1. Nepal: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2005/06–2025/26

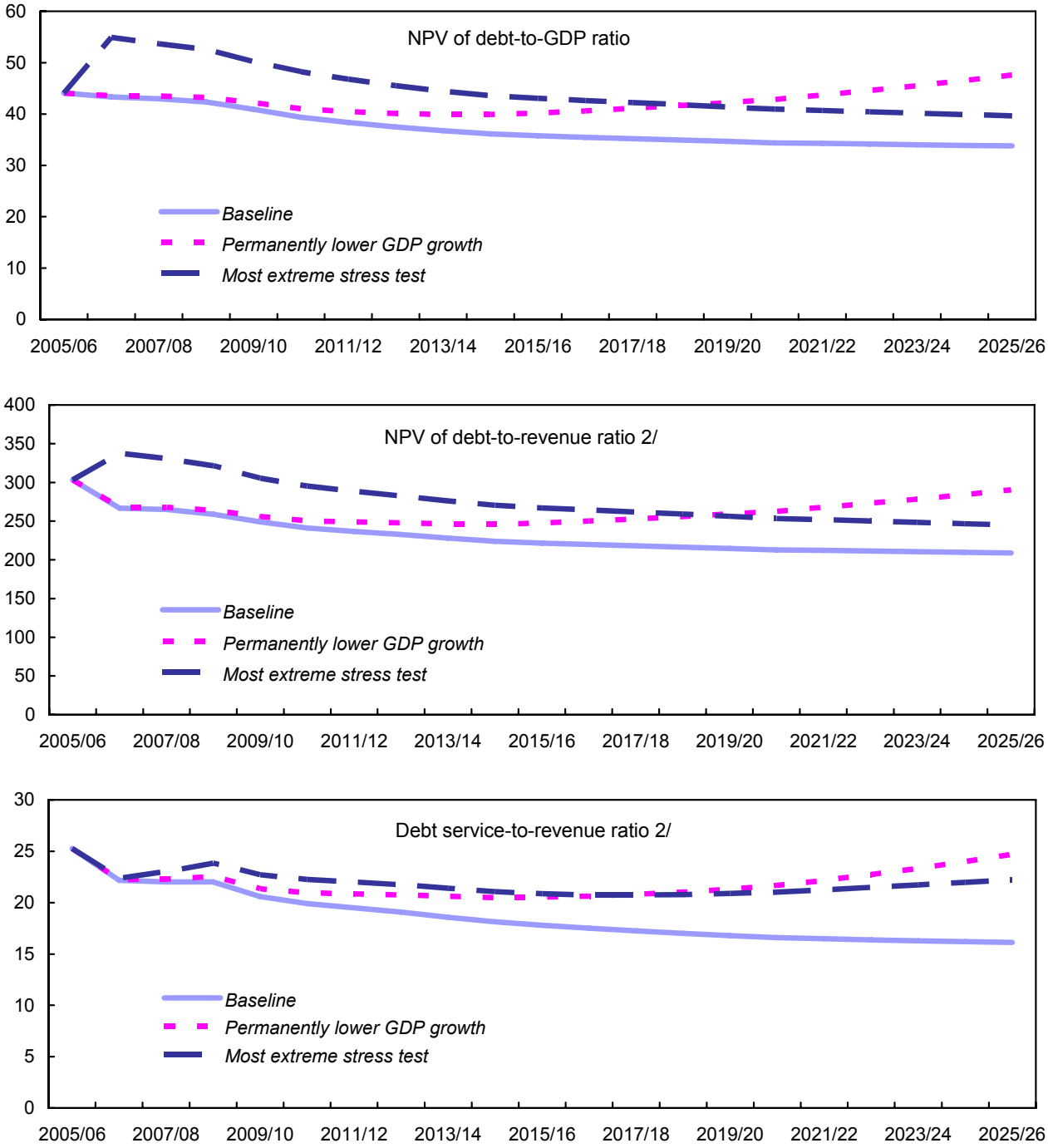
(In percent)



Source: Staff projections and simulations.

Figure I.2. Nepal: Indicators of Public Debt Under Alternative Scenarios, 2005/06–2025/26 1/ 2/

(In percent)



Source: Staff projections and simulations.

1/ Most extreme stress test is test that yields highest ratio in 2016.
2/ 2005 refers to fiscal year 2004/05.
3/ Revenue including grants.

Table I.1. Nepal: External Debt Sustainability Framework, Baseline Scenario, 2002/03– 2025/26 1/

(In percent of GDP, unless otherwise indicated)

	2002/03	2003/04	2004/05	Average 6/	Deviation 6/	Estimate 2005/06	Projections 2006/07	2007/08	2008/09	2009/10	2010/11	Average 2006/07–2010/11	2015/16	2025/26	Average 2011/12–2025/26
External debt (nominal) 1/	52.6	51.2	46.2			43.3	41.1	40.6	39.6	38.9	38.5		35.1	28.1	
Of which : public and publicly guaranteed (PPG)	49.3	48.3	43.4			40.4	38.1	37.5	36.3	35.6	35.2		31.6	24.5	
Change in external debt	-0.8	-1.4	-4.9			-2.9	-2.2	-0.5	-1.0	-0.7	-0.4		-0.8	-0.4	
Identified net debt-creating flows	-6.1	-9.7	-6.8			-3.1	-3.9	-3.8	-3.6	-2.7	-1.9		-0.2	0.2	
Non-interest current account deficit	-3.0	-3.4	-2.6	-2.2	3.1	-2.7	-3.0	-2.2	-1.8	-0.8	0.0	-1.8	1.6	2.2	1.8
Deficit in balance of goods and services	13.8	13.8	16.5			20.1	21.8	22.5	22.6	22.4	22.0		20.0	16.7	
Exports	16.9	18.0	16.1			15.3	15.5	16.0	16.2	16.5	16.7		17.1	18.3	
Imports	30.7	31.8	32.6			35.4	37.4	38.5	38.8	38.9	38.7		37.1	34.9	
Net current transfers (negative = inflow)	-16.5	-17.1	-18.3	-13.5	4.0	-21.6	-23.6	-23.4	-23.1	-21.8	-20.6	-22.4	-17.3	-13.7	-16.1
Other current account flows (negative = net inflow)	-0.3	-0.1	-0.7			-1.2	-1.3	-1.3	-1.4	-1.4	-1.4		-1.1	-0.7	
Net FDI (negative = inflow)	-0.2	0.0	0.0	-0.1	0.1	0.1	0.0	-0.2	-0.2	-0.3	-0.3	-0.1	-0.4	-0.8	-0.5
Endogenous debt dynamics 2/	-2.9	-6.3	-4.2			-0.4	-0.9	-1.4	-1.6	-1.7	-1.6		-1.5	-1.2	
Contribution from nominal interest rate	0.4	0.5	0.4			0.4	0.3	0.3	0.3	0.3	0.4		0.4	0.3	
Contribution from real GDP growth	-1.7	-1.7	-1.3			-0.8	-1.2	-1.7	-1.9	-2.0	-2.0		-1.8	-1.5	
Contribution from price and exchange rate changes	-1.7	-5.0	-3.3			…	…	…	…	…	…		…	…	
Residual (3–4) 3/	5.3	8.3	1.8			0.2	1.8	3.3	2.6	2.0	1.5		-0.6	-0.7	
Of which : exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
NPV of external debt 4/	...	...	29.3			27.8	26.8	26.8	26.4	26.3	26.2		24.7	19.5	
In percent of exports	...	...	181.6			181.9	172.3	167.7	163.0	159.7	157.4		144.0	106.8	
NPV of PPG external debt	...	...	26.4			24.9	23.7	23.6	23.2	23.0	22.9		21.2	15.9	
In percent of exports	...	...	163.8			162.9	152.8	147.9	143.0	139.5	137.2		123.8	87.3	
Debt service-to-exports ratio (in percent)	9.6	9.7	9.4			9.0	9.1	8.7	8.5	8.4	8.1		8.9	7.5	
PPG debt service-to-exports ratio (in percent)	9.6	9.7	9.4			9.0	9.1	8.7	8.5	8.4	8.1		8.9	7.5	
Total gross financing need (in billions of U.S. dollars)	-0.1	-0.1	-0.1			-0.1	-0.1	-0.1	-0.1	0.0	0.1		0.5	0.9	
Non-interest current account deficit that stabilizes debt ratio	-2.2	-2.0	2.4			0.2	-0.9	-1.7	-0.8	-0.1	0.4		2.5	2.7	
Key macroeconomic assumptions															
Real GDP growth (in percent)	3.3	3.8	2.7	3.9	2.0	1.9	3.0	4.5	5.0	5.5	5.5	4.2	5.5	5.5	5.5
GDP deflator in US dollar terms (change in percent)	3.3	10.5	6.9	1.5	4.9	6.9	5.9	2.2	3.6	3.1	3.1	4.1	1.2	0.7	1.1
Effective interest rate (percent) 5/	0.9	1.0	0.9	1.0	0.2	0.8	0.8	0.8	0.9	0.9	1.0	0.9	1.2	1.2	1.2
Growth of exports of G&S (U.S. dollar terms, in percent)	6.0	22.2	-1.7	1.9	19.0	3.2	11.4	9.7	10.3	10.4	10.1	9.2	7.5	6.8	7.4
Growth of imports of G&S (U.S. dollar terms, in percent)	6.0	18.7	12.6	5.4	14.6	18.1	15.8	10.0	9.8	8.9	8.2	11.8	5.9	5.6	6.0
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	50.6	43.8	40.6	41.6	39.1	40.1	42.7	39.3	44.3	41.4
Memorandym item:															
Nominal GDP (billions of U.S. dollars)	5.9	6.7	7.4			8.1	8.8	9.4	10.2	11.1	12.1		17.1	32.1	

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table I.2. Nepal: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2005/06–2025/26

(In percent)

	Estimate	Projections						
	2005/06	2006/07	2007/08	2008/09	2009/10	2010/11	2015/16	2025/26
				NPV of debt-to-GDP ratio				
Baseline	25	24	24	23	23	23	21	16
A. Alternative scenarios								
A1. Key variables at their historical averages in 2007–26 1/	25	26	28	29	29	30	25	7
A2. New public sector loans on less favorable terms in 2007–26 2/	25	24	25	25	26	26	27	26
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007–08	25	24	24	24	24	24	22	17
B2. Export value growth at historical average minus one standard deviation in 2007–08 3/	25	26	30	29	29	28	26	18
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2007–08	25	26	27	27	27	26	25	19
B4. Net nondebt creating flows at historical average minus one standard deviation in 2007–08 4/	25	32	40	39	38	38	33	21
B5. Combination of B1-B4 using one-half standard deviation shocks	25	35	49	47	46	45	39	24
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	25	33	33	32	32	32	30	22
				NPV of debt-to-exports ratio				
Baseline	163	153	148	143	140	137	124	87
A. Alternative scenarios								
A1. Key variables at their historical averages in 2007–26 1/	163	169	172	177	179	179	146	41
A2. New public sector loans on less favorable terms in 2007–26 2/	163	157	156	155	156	158	160	142
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007–08	163	152	147	142	139	137	123	87
B2. Export value growth at historical average minus one standard deviation in 2007–08 3/	163	225	334	321	311	303	266	173
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007–08	163	152	147	142	139	137	123	87
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007–08 4/	163	207	252	241	232	224	191	113
B5. Combination of B1-B4 using one-half standard deviation shocks	163	256	389	372	357	345	293	169
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	163	152	147	142	139	137	123	87
				Debt service ratio				
Baseline	9	9	9	9	8	8	9	8
A. Alternative scenarios								
A1. Key variables at their historical averages in 2007–26 1/	9	10	10	10	10	10	12	8
A2. New public sector loans on less favorable terms in 2007–26 2/	9	9	9	9	9	9	10	10
B. Bound tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007–08	9	9	9	9	8	8	9	7
B2. Export value growth at historical average minus one standard deviation in 2007–08 3/	9	12	16	17	16	16	19	15
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2007–08	9	9	9	9	8	8	9	7
B4. Net nondebt creating flows at historical average minus one standard deviation in 2007–08 4/	9	9	10	11	11	10	14	10
B5. Combination of B1-B4 using one-half standard deviation shocks	9	11	15	16	16	15	22	15
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	9	9	9	9	8	8	9	7
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	40	40	40	40	40	40	40	40

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and nondebt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline, while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers, FDI and capital flows in the financial account.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table I.3. Nepal: Public Sector Debt Sustainability Framework, Baseline Scenario, 2002/03–2025/26

(In percent of GDP, unless otherwise indicated)

	Actual			Historical	Standard	Estimate	Projections					2006/07–	Projections		2011/12–
	2002/03	2003/04	2004/05	Average 2/	Deviation 2/	2005/06	2006/07	2007/08	2008/09	2009/10	2010/11	2010/11 Average	2015/16	2025/26	2025/26 Average
Public sector debt 1/	67.2	65.4	59.5			56.7	54.7	53.8	52.5	50.4	48.5		42.8	38.9	
Of which: foreign-currency denominated	49.3	48.3	43.4			40.4	38.1	37.5	36.3	35.6	35.2		31.6	24.5	
Change in public sector debt	-0.7	-1.8	-5.9			-2.8	-2.0	-0.8	-1.4	-2.0	-1.9		-0.8	-0.2	
Identified debt-creating flows	-5.4	-4.8	-6.0			-0.9	-1.7	-1.6	-0.7	-2.0	-0.7		-0.1	0.4	
Primary deficit	0.1	-0.3	-0.3	2.1	1.7	0.6	1.7	1.8	1.8	1.8	1.8	1.6	1.8	2.0	1.9
Revenue and grants	14.4	14.4	15.6			14.5	16.2	16.2	16.3	16.4	16.3		16.1	16.2	
Of which: grants	2.1	2.3	2.7			2.3	3.0	2.8	2.7	2.6	2.5		2.3	2.3	
Primary (noninterest) expenditure	14.5	14.1	15.3			15.2	18.0	18.0	18.2	18.2	18.1		18.0	18.2	
Automatic debt dynamics	-5.6	-4.5	-5.7			-1.5	-3.5	-3.4	-2.6	-3.8	-2.5		-1.9	-1.6	
Contribution from interest rate/growth differential	-2.4	-2.9	-1.7			-1.3	-1.8	-2.2	-2.3	-2.5	-2.4		-1.9	-1.7	
Of which: contribution from average real interest rate	-0.2	-0.8	0.0			-0.2	-0.1	0.1	0.2	0.2	0.2		0.4	0.4	
Of which: contribution from real GDP growth	-2.2	-2.2	-1.6			-1.1	-1.7	-2.3	-2.6	-2.7	-2.6		-2.3	-2.0	
Contribution from real exchange rate depreciation	-3.1	-1.5	-4.0			-0.2	-1.7	-1.2	-0.2	-1.3	-0.1		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g., bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	4.7	3.0	0.1			-1.9	-0.3	0.8	-0.6	0.0	-1.2		-0.7	-0.5	
Other Sustainability Indicators															
NPV of public sector debt	17.9	46.6	45.5			44.1	43.3	43.2	42.5	41.1	39.5		35.9	33.9	35.4
o/w foreign-currency denominated	0.0	29.5	29.3			27.8	26.8	26.8	26.4	26.3	26.2		24.7	19.5	22.9
o/w external	...	29.5	29.3			27.8	26.8	26.8	26.4	26.3	26.2		24.7	19.5	22.9
NPV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0			...	...	...	...	...	...		...	...	
Gross financing need 2/	3.7	3.2	3.4			4.3	5.3	5.4	5.5	5.2	5.0		4.7	4.6	4.7
NPV of public sector debt-to-revenue ratio (in percent) 3/	124.3	322.8	291.4			302.9	266.8	266.5	260.3	250.1	242.1		222.4	209.3	219.1
o/w external	...	204.4	187.8			191.2	164.9	165.4	161.7	159.9	160.6		152.9	120.6	142.1
Debt service-to-revenue ratio (in percent) 3/ 4/	24.6	24.2	23.7			25.3	22.2	22.0	22.1	20.6	19.9		17.8	16.1	17.3
Primary deficit that stabilizes the debt-to-GDP ratio	0.9	1.5	5.5			3.5	3.8	2.6	3.2	3.8	3.7		2.6	2.1	2.5
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	3.4	3.4	2.5	3.9	2.0	1.9	3.0	4.5	5.0	5.5	5.5	4.2	5.5	5.5	5.5
Average nominal interest rate on forex debt (in percent)	0.9	1.0	0.9	1.1	0.2	1.0	1.0	1.1	1.1	1.2	1.2	1.1	1.4	1.3	1.4
Average real interest rate on domestic currency debt (in percent)	2.0	-0.1	0.0	0.8	1.9	-1.6	-0.7	0.4	1.3	1.0	1.1	0.2	2.8	2.1	2.4
Real exchange rate depreciation (in percent, + indicates depreciation)	-6.4	-3.3	-8.5	0.1	7.4	-0.4	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	4.5	5.2	4.8	5.3	2.0	7.2	7.2	6.0	5.5	5.0	5.0	6.0	3.5	3.0	3.5
Growth of real primary spending (deflated by GDP deflator, in percent)	-5.0	0.7	10.8	4.8	8.9	1.0	22.1	4.6	6.1	5.8	4.6	7.4	5.7	5.6	5.5

Sources: Nepalese authorities; and Fund staff estimates and projections.

1/ Public and publicly guaranteed debt.

2/ Historical averages and standard deviations are derived over the past 10 years.

Table I.4. Nepal: Sensitivity Analysis for Key Indicators of Public Debt, 2005/06–2025/26

	Projections							
	2005/06	2006/07	2007/08	2008/09	2009/10	2010/11	2015/16	2025/26
	NPV of debt-to-GDP ratio							
Baseline	44	43	43	43	41	40	36	34
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	44	43	44	44	43	42	42	44
A2. Primary balance is unchanged from 2006	44	42	41	39	37	34	26	15
A3. Permanently lower GDP growth 1/	44	44	44	43	42	41	40	48
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2007–2008	44	44	45	45	44	43	41	43
B2. Primary balance is at historical average minus one standard deviations in 2007–2008	44	45	47	46	44	43	39	36
B3. Combination of B1-B2 using one half standard deviation shocks	44	45	46	45	44	42	38	35
B4. One-time 30 percent real depreciation in 2007	44	55	54	53	50	48	43	40
B5. 10 percent of GDP increase in other debt-creating flows in 2007	44	53	52	51	49	48	43	39
	NPV of debt-to-revenue ratio 2/							
Baseline	303	267	267	260	250	242	222	209
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	303	267	270	268	262	258	258	261
A2. Primary balance is unchanged from 2006	303	260	254	241	225	211	163	93
A3. Permanently lower GDP growth 1/	303	268	269	265	257	251	248	291
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2007–2008	303	270	279	275	267	262	254	263
B2. Primary balance is at historical average minus one standard deviations in 2007–2008	303	279	289	282	271	262	241	224
B3. Combination of B1-B2 using one half standard deviation shocks	303	274	284	276	265	256	233	214
B4. One-time 30 percent real depreciation in 2007	303	338	333	323	306	296	268	246
B5. 10 percent of GDP increase in other debt-creating flows in 2007	303	324	321	313	300	291	266	244
	Debt service-to-revenue ratio 2/							
Baseline	25	22	22	22	21	20	18	16
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	25	22	23	23	22	22	21	20
A2. Primary balance is unchanged from 2006	25	22	20	19	17	16	12	2
A3. Permanently lower GDP growth 1/	25	22	22	23	21	21	21	25
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2007–2008	25	22	23	24	23	22	21	22
B2. Primary balance is at historical average minus one standard deviations in 2007–2008	25	22	26	28	23	21	19	18
B3. Combination of B1-B2 using one half standard deviation shocks	25	22	25	26	22	21	18	17
B4. One-time 30 percent real depreciation in 2007	25	23	23	23	22	22	20	19
B5. 10 percent of GDP increase in other debt-creating flows in 2007	25	22	42	30	25	23	20	22

Sources: Nepalese authorities; and Fund staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.